

ROBOTIC HAND DESIGN FOR POSITION AND FORCE CONTROL

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Abstract. This paper studies the design of a hybrid control system, one that controls both position and force of a robot gripper with fingertip-like contacts. By attaching a strain gauge on the finger's flexural beam, a second-order system is analyzed for variable stiffness. This system's dynamics is further calculated after choosing the forward-path and velocity gains. An adaptive-control system for real-time operation is finally presented, where the robot hand may perform force- and position-controlled tasks independently.

1. Theoretical background

Our choice of a system to study is the use of a DC electric motor to position an inertial load through a gearbox. Such a system is very common in robot hand designs, where a stepper motor is usually commanded to position the entire arm, or a segment of it. This paper concentrates on the last link of the robot arm, namely the gripper motor, gearbox, and load (fig. 1).

Knowing the motor and load inertia J_m and J_l , and the friction coefficients c_m and c_l , the torque equations for the motor and load shafts are given in (1). Not all of the torque τ_m is available at the gearbox input shaft, since the armature inertia has to be overcome, as does the viscous friction in the armature bearings. Substituting the gearbox equations (2) in (1) yields the final relation between the motor torque τ_m and the motor angular speed ω_m , given in (3). As can be easily seen, the effect of the gearbox is to reduce both the load inertia and load shaft damping by a factor of N^2 , and therefore it is not uncommon to find gearbox ratios in the order of several thousands. Under these circumstances, it is the load inertia rather than the motor inertia that can be neglected, as it is also the case in the following design.

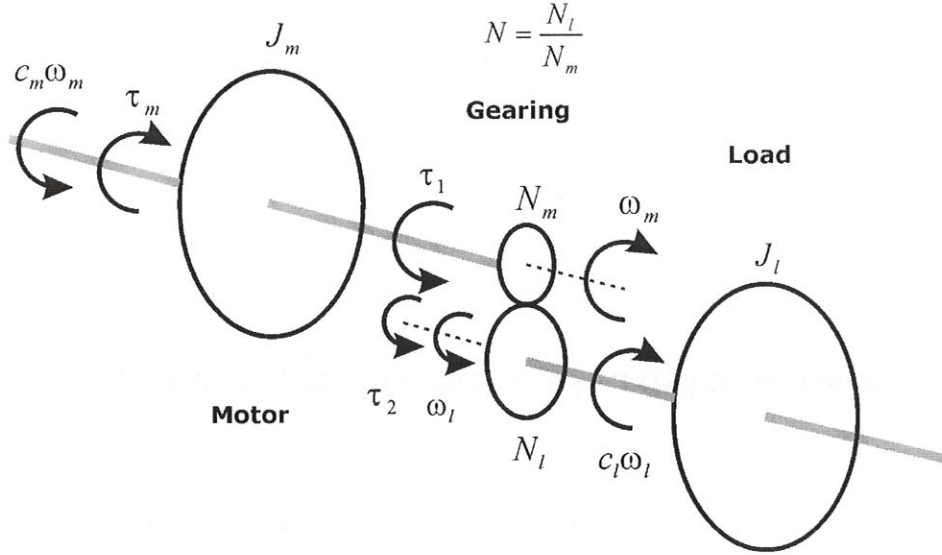


Fig. 1 Free-body diagram of motor, gearbox, and load

$$\tau_m - \tau_1 - c_m \omega_m = J_m \dot{\omega}_m \quad (1)$$

$$\tau_2 - c_l \omega_l = J_l \dot{\omega}_l$$

$$\omega_l = \omega_m / N \quad \tau_2 = N \tau_1 \quad (2)$$

$$\tau_m = \left(J_m + \frac{J_l}{N^2} \right) \dot{\omega}_m + \left(c_m + \frac{c_l}{N^2} \right) \omega_m \quad (3)$$

The transfer function, in complex frequency domain, is obtained by taking the Laplace transform of equation (3) and rearranging in the form (4):

$$\frac{\Omega}{\Gamma} = \frac{1}{c + sJ} \quad (4)$$

where J is the total inertia referred to the motor shaft and c is the total damping, also referred to the motor shaft. The block diagram for the complete positioning system is given in fig. 2. The time domain correspondents of the functions Ω and Γ are ω_m and τ_m , respectively. The input angular position θ_i is passed through several blocks in order to obtain the desired load angular position θ_o , using a closed loop. In operation, the set point is generated by means of a potentiometer or, even better, a digital to analog converter in the case of a computerized command. The voltage that is obtained after passing this block's transfer function K_p is compared to the achieved position of the load, measured typically by a resolver to digital converter. If both blocks have the same transfer function, they can be represented by a single block, after the junction (see fig. 2). The resulted error voltage v_e is passed through the amplifier K_a since the motor may require greater power levels than those obtained directly from the summing junction. For simplicity, the motor electrical circuit

is considered a resistance R in series with an inductance L . The relationship between the applied voltage v_a and the current i is given by

$$v_a = iR + L \frac{di}{dt} \quad (5)$$

and taking its Laplace transform yields

$$\frac{I}{V_a} = \frac{1}{R + sL} \quad (6)$$

Because in practice the internal inductivity is much smaller than the resistance, in the motor transfer function the term sL will be neglected from now on. The motor torque τ_m is obtained by the field current that establishes a magnetic flux that interacts with the armature field, and this phenomenon is represented by the motor torque constant K_m . An additional velocity feedback K_b has been added to adjust the damping of the response. It is particularly important to design for not less than critical damping so that no overshoot occurs in the position loop. This makes possible, with the same system, the command of the gripper for two extreme tasks: the grasping of very light and delicate objects, and of rigid ones. The load angular speed ω_l is referred to the motor speed ω_m by the reduction gear ratio. The closed-loop transfer function of the control system presented in fig. 2 is given in (7).

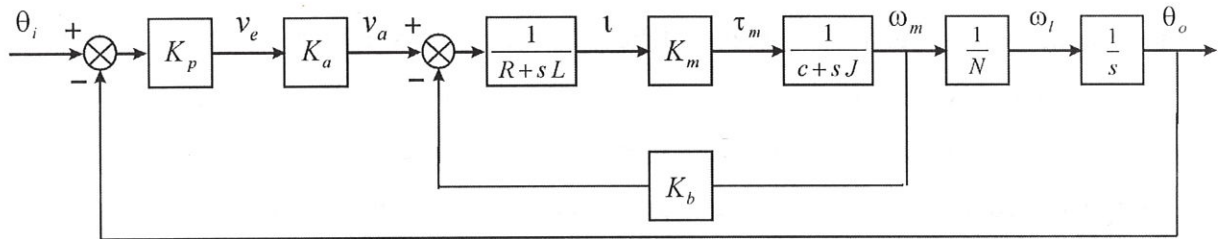


Fig. 2 Position control system block diagram

$$\frac{\theta_o}{\theta_i} = \frac{K_p K_a K_m / NRJ}{s^2 + s(cR + K_m K_b) / RJ + K_p K_a K_m / NRJ} \quad (7)$$

As long it is a second order system, it might be also expressed as a function of the undamped natural circular frequency ω_n and the damping ratio ζ :

$$\frac{\theta_o}{\theta_i} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (8)$$

Combining relations (7) and (8) yields

$$\omega_n = \sqrt{K_p K_a K_m / NRJ} \quad (9)$$

$$\zeta = \sqrt{N / RJ K_p K_a K_m} (cR + K_m K_b) / 2 \quad (10)$$

Knowing that

$$\omega_n = \alpha_s / T_s \zeta \quad (11)$$

where α_s is a factor that limits the transient response function within a corridor no wider than a desired value (e.g. for $\leq 1\%$ $\alpha_s = 4.6$) and T_s is the system time constant, after combining (9) - (11) the following condition must be imposed in order to have valid physical values for the supplementary velocity feedback:

$$T_s \leq 2J\alpha_s / c \quad (12)$$

and the relation for its calculation is

$$K_b = (2J\zeta\omega_n - c)R / K_m \quad (13)$$

The relation for obtaining the gain is finally given by

$$K_a = (cR + K_m K_b)^2 N / 4RJK_p K_m \quad (14)$$

2. Independent force control and position control

Up to this point, the reacting force that appears on the robot gripper is not taken into consideration. The robot can grasp any kind of objects and uses the same force in any situations. For some tasks, such as picking delicate objects, this system's design can be considered inappropriate because the gripper might damage the objects. To improve this, in fig. 3 a torque sensor is introduced as follows. The fingertip is connected to the last finger link by means of a cantilever beam to which a strain gauge is attached. The applied torque will be therefore computed from the strain gauge signal, and for each task that the hand performs, the appropriate finger stiffness has to be known. For example, picking up an egg would require a lower stiffness as compared to grasping a hammer.

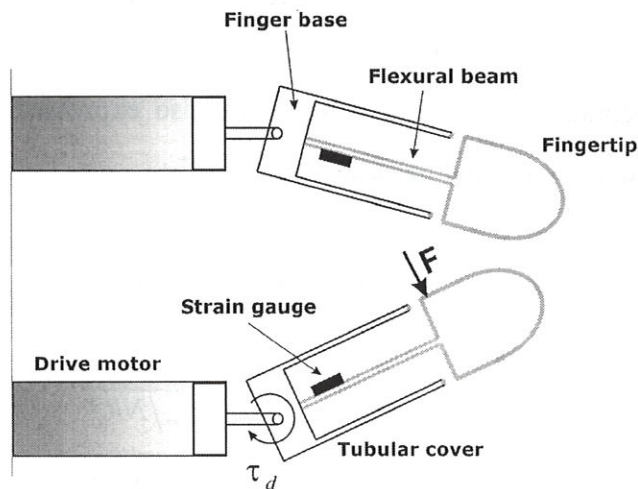


Fig. 3 Torque sensor for fingertip-like contacts

Since the force that deforms the cantilever is equivalent to a moment τ_d about the finger axis, it might be regarded as a disturbing torque, as shown in fig. 3. Because it is applied to the finger shaft – on the load side of the gearbox (see fig. 1) – the torque exerted on the motor shaft is τ_d/N . The resulting control system, with this disturbance torque considered, is presented in fig. 4 and the output angular position is given by (15).

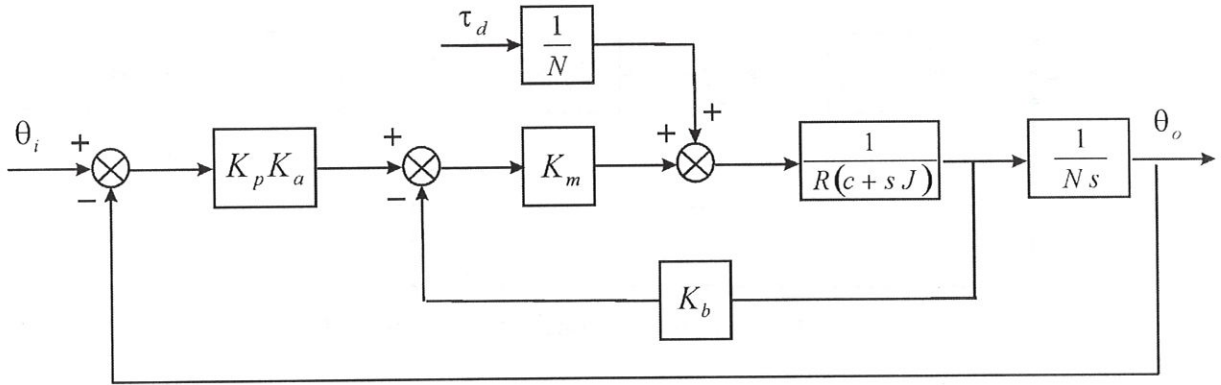


Fig. 4 Position and control system with disturbance

$$\theta_o = \frac{\theta_i K_p K_a K_m / NRJ + \tau_d / N^2 RJ}{s^2 + s(cR + K_m K_b) / RJ + K_p K_a K_m / NRJ} \quad (15)$$

Considering $\theta_i = 0$ and τ_d to be a unit step, the steady-state angular rotation of the finger link will be

$$\theta_o = 1 / N K_p K_a K_m \quad (16)$$

and for the numerical values $K_p = 5.7$ [V/rad], $K_m = 0.2$ [Nm/V], $c = 2\text{E-}5$ [Nms/rad], $J = 1.75\text{E-}5$ [Nms²/rad], $R = 10$ [Ω], $N = 1000$, $\alpha_s = 4.6$, $\zeta = 1$, $T_s = 8$ [s] an angular rotation of 0.99° is obtained (the previous numerical values are used in all following calculations, except when otherwise not specified). This is the natural stiffness of the servo system and is the stiffness that would be felt if the output shaft were grasped and turned. The potentiometer would detect the rotation and generate a torque to restore the position of the shaft to that defined by θ_i . The question is how to change this stiffness so that if the finger closes a small amount, large interaction forces are not generated. The stiffness is expressed by (17).

$$S = \tau_d / \theta_o \quad (17)$$

It seems that with the existing system configuration it will not be possible to design for both position and force control simultaneously. The problem can be solved if the torque τ_d can be measured in some way, and this signal is further used to drive the output to a larger value than

would otherwise be achieved. In fig. 5, the signal from the torque transducer is multiplied by some gain E and added to the feedback loop at the first summing junction. If a unit torque is applied, the system would respond with a rotation of 0.99° as before, except this time an additional voltage causes a larger rotation θ_o . For this system, the output is given by (18).

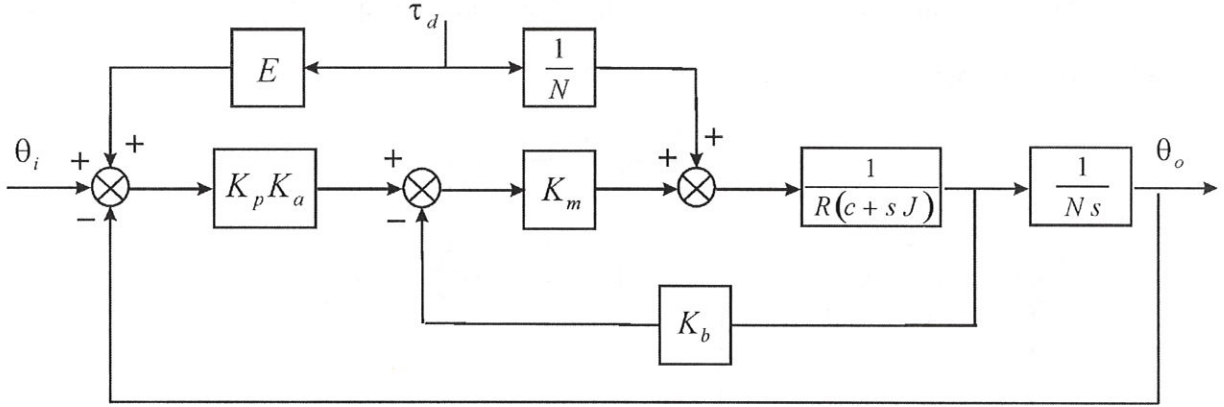


Fig. 5 Force feedback for variable stiffness

$$\theta_o = \frac{\theta_i K_p K_a K_m / NRJ + (1 + NEK_p K_a K_m) \tau_d / N^2 RJ}{s^2 + s(cR + K_m K_b) / RJ + K_p K_a K_m / NRJ} \quad (18)$$

Again considering $\theta_i = 0$ and applying the final-value theorem for a unit step torque input, one obtains

$$\theta_o = \tau_d (1 + ENK_p K_a K_m) / NK_p K_a K_m \quad (19)$$

and substituting (19) in (17) yields the dependence of the gain E with the joint stiffness. The behavior of (20) is illustrated in fig. 6.

$$E(s) = (NK_p K_a K_m - s) / sNK_p K_a K_m \quad (20)$$

Considerable information may be obtained regarding the nature of response of a second-order system by studying its behavior when subject to a step input. In such a case, equation (8) indicates that the output is given by

$$H(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} \quad (21)$$

and applying the inverse Laplace transform, for critically damped response ($\zeta = 1$), yields

$$h(t) = 1 - (1 + \omega_n t) \exp(-\omega_n t) \quad (22)$$

Figure 7 shows the step response for various values of the system time constant, T_s . The highest value considered for T_s (2.97 s) is deduced later in this paper, and is the new higher limit from now

on (according to (12), T_s initially was limited to 8.05 [s]). One important observation that can be drawn out is that the step response given by (22) has a numerical value ≈ 0.95 for $t = T_s$.

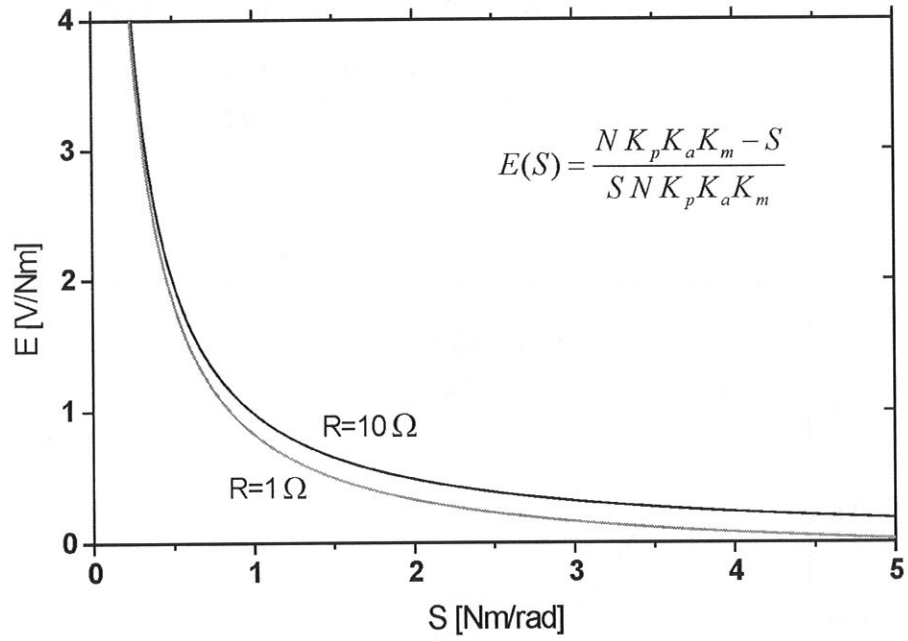


Fig. 6 Strain gauge gain variation with joint stiffness

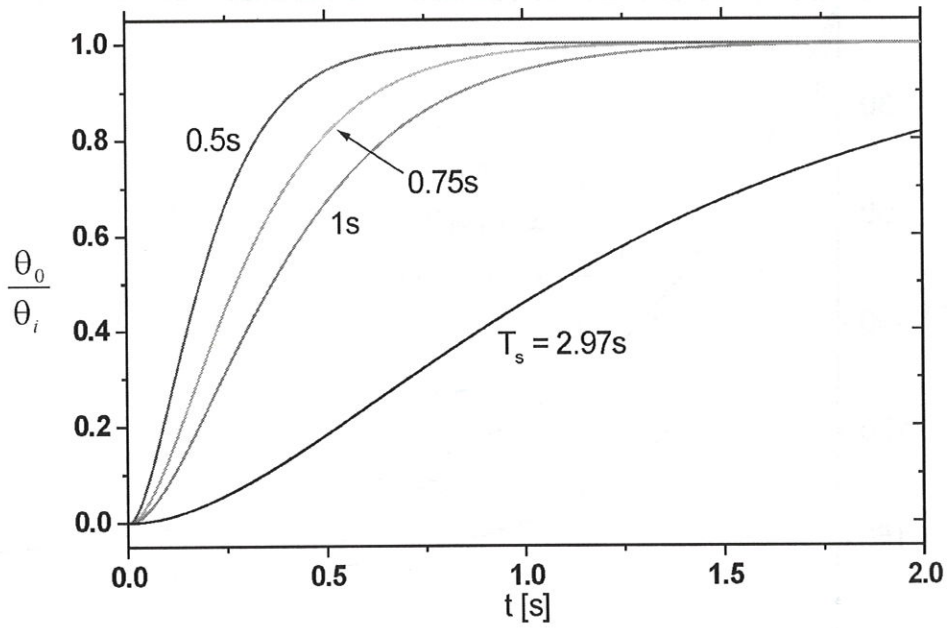


Fig. 7 Step response for varying T_s

Applying the transformation $s = j\omega$ in (18) both modulus and phase for the transfer function can be obtained. Their expressions are given by (23) and (24).

$$\left| \frac{\theta_o}{\theta_i} \right| = \frac{K_p K_a K_m}{\sqrt{(K_p K_a K_m - \omega^2 NRJ)^2 + [\omega N(cR + K_m K_b)]^2}} \quad (23)$$

$$\Phi = \text{atan} \frac{-\omega N(cR + K_m K_b)}{K_p K_a K_m - \omega^2 NRJ} \quad (24)$$

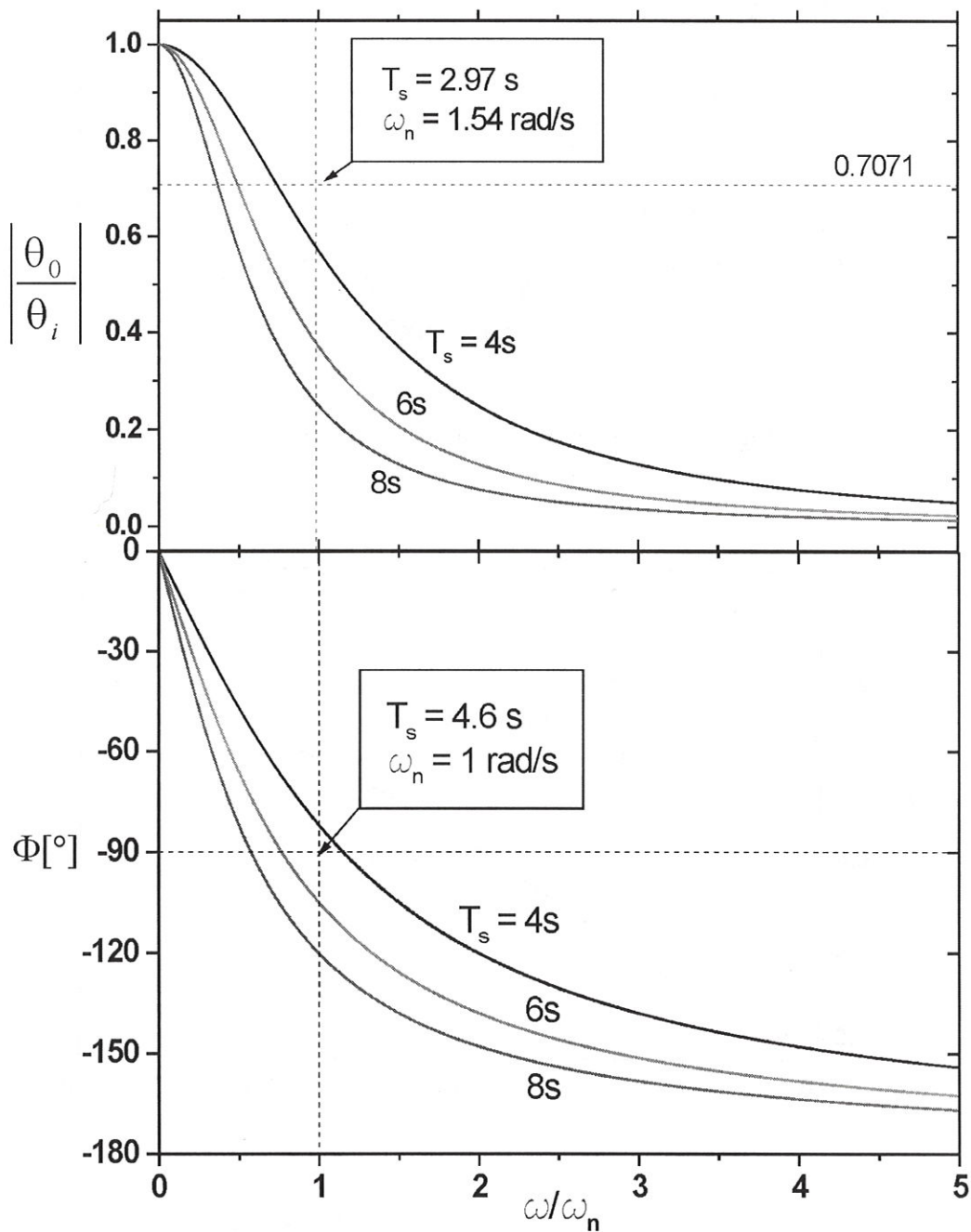


Fig. 8 Modulus and phase of the transfer function

An important feature that has to be carried out by the system is that the modulus of the complex transfer function is greater than $1/\sqrt{2} \approx 0.7071$. Values for T_s smaller than 2.97 [s] have resulted after numerically solving equation (23) with $\omega = \omega_n$, and the phase equals $-\pi/2$ for $T_s = \alpha_s$ under the same circumstances (see fig. 8). Having in mind that $T_s < 2.97$ [s], the plots for the position gain K_a , velocity feedback gain K_b , and output position θ_o were calculated and plotted in fig. 9.

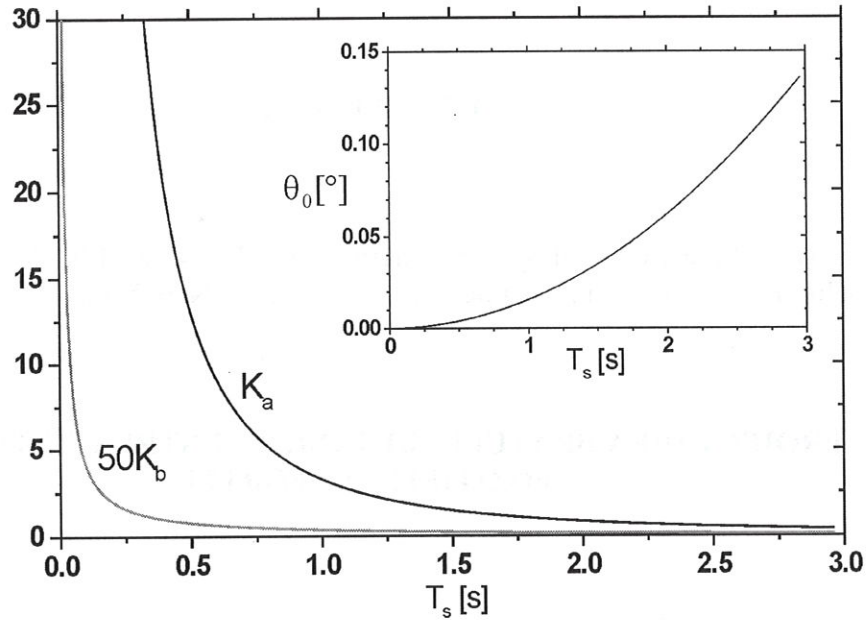


Fig. 9 Position (K_a) and velocity feedback (K_b) gain, and output positioning θ_o as a function of T_s

After analyzing fig. 9 some observations regarding the system parameters have to be done. First, the output θ_o rises exponentially from zero. As can be seen from fig. 7, the lower the system time constant T_s , the faster the time response. It would be the right choice to look for the lowest physically T_s possible, if it wouldn't be another limiting factor, namely the potentiometer-block. It is obvious that the highest precision is reached when using a high-resolution digital to analog converter (DAC). Considering M bits to convert and a $\Delta\theta$ desired angular precision, the minimum number of bits for the DAC is obtained after

$$M > \lceil \log_2 360/\Delta\theta \rceil \quad (25)$$

and for $\Delta\theta = 0.01^\circ$ ($T_s \approx 0.8$ [s]) results $M \geq 16$. Such resolutions are above average DAC performances, therefore in practice should be used a value of M around 12. In this latter case, the corresponding $\Delta\theta$ is 0.087° and $T_s \approx 2.37$ [s] (see fig. 9). This accuracy problem can be overshoot if the DAC is designed to command less than 2π radians. For example, a 1 radian arc combined

with a 12-bit DAC gives $\Delta\theta = 0.014^\circ$. This case has a correspondent $T_s = 0.95$ [s], $K_a = 3.599164$, $K_b = 7.473684E-3$, and $\omega_n = 4.842105$ [rad/s].

3. Conclusions

A hybrid control system, for both position and force control, and which can be used by a robot gripper, was analyzed. Starting from the transfer function in complex frequency domain, and continuing with the time response to unit step input, the transfer block coefficients were calculated numerically and the results were analyzed for a practical servo-system.

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PROIECTAREA BRATULUI DE ROBOT PENTRU CONTROLUL POZITIEI SI AL FORTEI

Rezumat. Aceasta lucrare studiaza proiectarea unui sistem de control mixt, avand comanda simultana a pozitiei si fortei pentru dispozitivul de prindere al unui robot, asemanator cu degetele umane. Atasand un traductor de tensionare pe tija de incovoiere a degetului, se obtine un sistem de ordinul doi care este analizat pentru diferite rigiditati de prindere. Odata cunoscute castigul pentru viteza si traiectoria de urmat, dinamica acestui sistem este calculata si analizata. In final se prezinta un sistem de control adaptiv, pentru lucrul in timp real, in care dispozitivul de prindere al robotului poate indeplini sarcini cu controlul independent al pozitiei si fortei de prindere.

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